

- 3 A particle P is moving in a horizontal straight line. Initially P is at the point O on the line and is moving with velocity 25 ms^{-1} . At time $t \text{ s}$ after passing through O , the acceleration of P is $\frac{4000}{(5t+4)^3} \text{ ms}^{-2}$ in the direction PO . The displacement of P from O at time t is $x \text{ m}$.

Find an expression for x in terms of t .

[5]

[illegible]